Enrolment Number										

Total No. of printed pages = 04

Winter, 2026

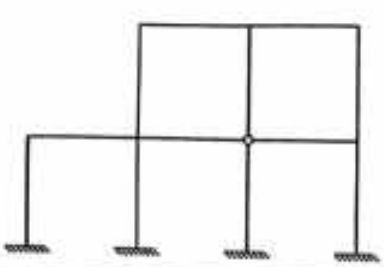
B. Tech 4th Semester Examination

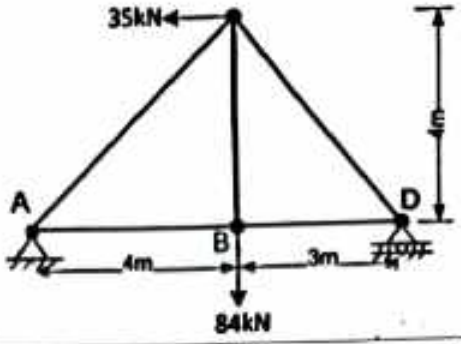
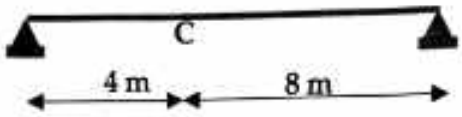
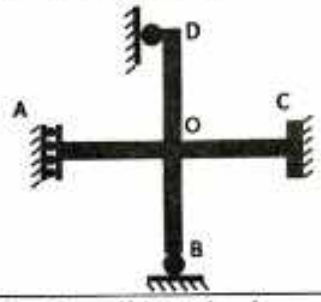
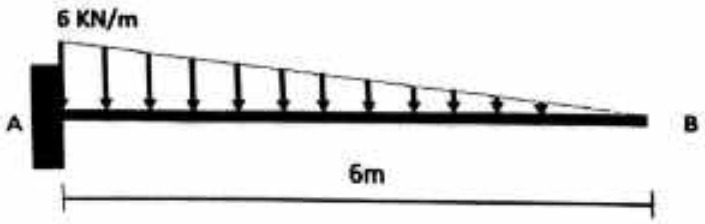
Structural Analysis – Theory

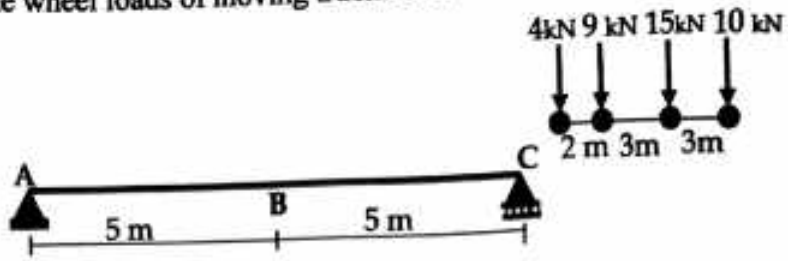
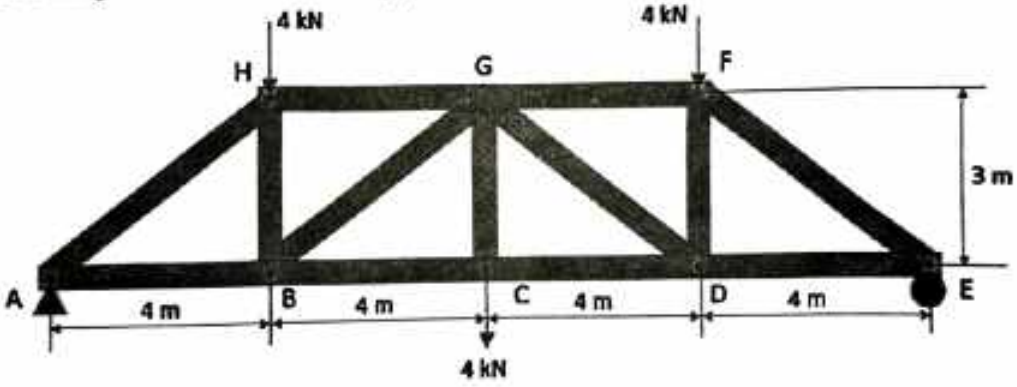
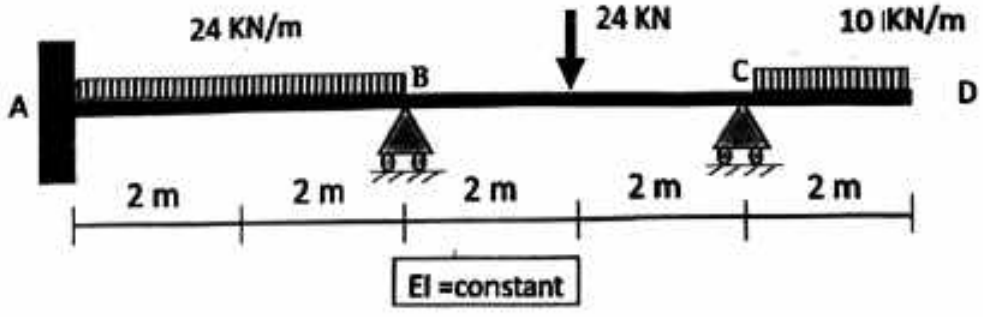
Course Code: BCE23405T

Full Marks: 50 **Time: 2 hours**

The figure in the margin indicates full marks for the questions.

S. No	Questions	Marks
	Section 1 [10 Marks]	
1a	1.If the degree of static indeterminacy of a structure is zero, the structure is called	1
1b	Influence lines are especially useful for structures subjected to .....loads.	1
1c	What do you mean by stiffness of a member.	1
1d	Define distribution factor.	1
1e	Rankine's formula combines.....and Euler theory.	1
1f	Explain principal of superposition.	1
1g	The effective length of a cantilever column is	1
1h	Castigliano's theorem is applicable to structures behaving within the..... range.	1
1i	The fixed end moment for a beam carrying a point load at the midspan is .....at both ends.	1
1j	The method of joints is mainly used to determine.....	1
	Section 2 [8 Marks] Answer any 4 questions [2 marks each]	
2a	Determine the degree of indeterminacy for the rigid-jointed frames indicated in Figures. 	2

2b	Classify the truss in the figure below as stable, unstable, statically determinate, or static indeterminate. Also calculate the reactions of the given truss. 	2
2c	A three hinged parabolic arch of uniform cross section of span 60m and rise 10m carries a uniformly distributed load of 10 kN/m over the entire span. Calculate the vertical reactions and the horizontal thrust at supports.	2
2d	Draw the influence line diagram for moment at point C of the given beam 	2
2e	Calculate the Distribution factor for each member of the beam shown in figure below. EI and L is constant for all the members. 	2
2f	What is the ratio of the strength of a solid steel column of 200mm diameter to that of a hollow circular steel column of the same cross-sectional area and a wall thickness of 20mm. The two columns have the same length and same boundary condition.	2
	Section 3 [12 Marks] Answer any 3 questions [4 marks each]	
3a	Derive the expression for crippling load for columns with one end fixed and other hinged.	4
3b	Determine the maximum deflection of the beam E = 200 GPa and I=100x10^6 mm^4.Use unit load method. 	4

3c	Determine the maximum positive shear created at point B in the beam shown in Figure due to the wheel loads of moving truck. Draw the influence line for shear. 	4
3d	In the analysis of portal frame ABCD with sway, calculate the Restoring force (R) at the top of the frame. The end column moments are as follows: M_{AB} = 9.58 kN-m M_{CD} = -15.00 kN-m M_{BA} = 19.34 kN-m M_{DC} = 0 kN-m The length of column AB and CD are 10 m and 15 m respectively.	4
	Section 4 [20 Marks] Answer any 2 questions [10 marks each]	
4a	Calculate the forces in each member of the truss using **method of joints**. Support A is hinged and E is a roller support. 	10
4b	Analyse the beam shown in Figure using **Slope deflection method** if support B sinks by 20 mm. Draw the bending moment diagrams for the beam. Take EI = $2x10^4$ kN-m^2 	10

Analyse the frame shown using **Moment distribution method**. Draw the bending moment diagrams for the frame. EI is constant.

